On the discussion questions regarding Engel's 'Logic, Reasoning and the Logical Constants' (December 17, 2025)

Here are a couple of thoughts and observations on the discussion questions. As always, comments are welcome!

All groups:
"What are reasons to consider truth and validity to be *normative* concepts (Engel, 224, 233)? Is logical pluralism problematic from that perspective?"

One reason to regard truth and validity as normative concepts is related to the function of representations. A representation is supposed to be truthful: that is what distinguishes a representation from a mere sign. To put it differently, when we use a sign to represent, we intend there to be a specific and in principle verifiable relation between the sign and the signified: *this* sign signals the presence of *that* situation (or object, or whatever the sign is used to represent). Arbitrary use of signs is useless: if the relation between sign and signified were arbitrary, we would in effect not be able to use signs to represent and communicate. Representations are supposed to be true; if systematic truthfulness were not the norm, there would be no point in representing (saying, arguing about) anything.

This fundamental principle has been stated in the literature in different ways. A well-known example is David Lewis' "Convention of Truthfulness and Trust".[1] Lewis proposes that in order for a system of signs to function as a language, two things need to be in place: speakers use the signs to say what they believe to be true; and hearers trust speakers to do so and interpret them accordingly. Of course, exceptions exist: people lie, tell jokes, are sarcastic or ironic. But the basic convention is that of truthfulness: without that as a norm language cannot function. A similar idea can be found in Grice's 'Logic and Conversation' (see chapter 27 in the course book): the first conversational maxim that Grice formulates is the 'Maxim of Quality", which requires speakers to say only what they believe to be true. And Donald Davidson makes the concept of error, which is of course intimately related to that of truth, the distinctive characteristic of belief: only creatures that have the concept of error can be said to have beliefs.[2]

What is interesting to note is that considerations such as these seem to apply *directly* at the level of what humans do when using language, when reasoning, when planning to act, and so on. That suggests that a proper description of human cognitive activities has to take into account the fact that such activities are inherently subject to normative constraints. In other words, normativity is not something that is imposed from the outside, it is 'baked into' these very concepts.

The above only pertains to the concept of truth. But it seems plausible that similar considerations apply to that of validity. When we reason, when we draw conclusions from observations, when we hypothetically explore possible effects of actions, and so on, we do so under similar normative constraints. Drawing an inference is not a matter of stating an arbitrary connection between sentences, the connections need to be compelling. For if they weren't, there would not be a point in considering them in the first place.

Then what about logical pluralism? Can there really be different conceptions of truth and validity? There is a rub here, definitely. Too much pluralism seems to negatively affect our capacities to communicate and cooperate. A certain measure of shared rationality, of shared

[1] See David K. Lewis, *Convention. A Philosophical Study*, Harvard University Press, Cambridge, Mass., 1969.
[2] See Donald Davidson, 'Rational Animals', *Dialectica*, vol. 36, 1982.

logic, seems necessary in order to provide a basis for that, but how much needs to be shared, and what exactly, is difficult to determine independently and absolutely. The nature and the complexity of the tasks and of the reasoning they involve appear to be important factors.

A last observation might help to put the 'dangers of pluralism' in perspective. Although they are different logics, it turns out that classical logic and intuitionistic logic are systematically related. One result is due to Glivenko.[3] We map each classical formula φ to its double negation ¬¬φ; then we can prove that φ is a classical tautology if and only if ¬¬φ is an intuitionistic tautology. A similar result for predicate logic was obtained by Gödel and, independently, by Gentzen.[4]

Group 1:
"Engel (228) disagrees with the view that if we are not following an *explicit* rule, we 'act blindly'. What kind of implicit knowledge could account for that?"

Engels uses the term 'tacit knowledge' in connection with this. That may be an implicit reference to Chomsky's conception of the tacit (innate) knowledge of language that competent speakers of a language are supposed to have. The case of logic and the case of language do indeed appear similar in some respects. We are competent speakers, at least of our first language, yet we have no explicit theory of our language (let alone one that is both explicit and correct), so *that* is not what we employ when we exercise our competence. Arguably we are less competent reasoners than we are language users, nevertheless, in reasoning too, be it correct or incorrect, we do employ some kind of knowledge. But again, what we employ is not an explicit theory. And note that also when we reason incorrectly, we do employ some kind of systematic rules, we do not just randomly link things to other things and call that 'inference'. Evidence for that comes from the fact that we can come up with what we regard as justifications, also for incorrect reasoning; that we can be persuaded that we reasoned incorrectly by counterexamples; and so on. So, irrespective of the correctness of the results, reasoning is a rule-guided activity. However, it seems quite intuitive that when we are engaged in actual reasoning processes we need not, and generally are not, aware of the rules that guide us.

Then how do the rules guide us, if it is not through our consciously applying them? The question is whether the use of the concepts of 'rule' and 'knowledge' in this context is very helpful. There is normativity, that seems clear. (Cf., also above, concerning the question about the normativity of the concepts of truth and validity.) But what guides us are not rules in the sense of rules that are defined explicitly, that can be written down, discussed, agree upon, and that can be applied in a conscious, intentional manner. It seems much more a matter of attunement to what regulates speech/action in ways that make that speech/action relevant and appropriate. That is a way of being guided that is not introspectively accessible, and in that sense is beyond justification. However, as Wittgenstein notes (in *Philosophical Investigations*, 289): "To use a word without a justification does not mean to use it wrongfully." There is a right and wrong about our use of language and, similarly, about our reasoning practices, that is *not* based on awareness of and access to explicit rules, but that nevertheless normatively constrains us.

Group 2:
"From a proof-theoretic point of view, what we manipulate are strings of signs, not representations. Does that fit the way we reason in everyday life?"

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[3] In two French-language papers dating from 1928 and 1929. An English translation of the 1929 paper can be found in Paolo Mancosu (ed), *From Brouwer to Hilbert. The Debate on the Foundations of Mathematics in the 1920s*, Oxford: Oxford University Press, 1998.
[4] See K. Gödel, 'Zur intuitionistischen Arithmetik und Zahlentheorie', *Ergebnisse eines mathematischen Kolloquiums*, vol. 4, 1933.

We should start out by noting that the strictly formalistic interpretation of the proof-theoretic view is closely related Hilbert's formalism in the philosophy of mathematics.[5] Mathematics, according to Hilbert, is the manipulation of strings using a set of manipulation rules. It is not about mathematical objects, properties and relations, i.e., it comes without any ontological commitment. The main challenge, as Hilbert set out in what is known as 'Hilbert's Program', [6] is to provide consistent, complete and decidable axiomatisations of all of mathematics. Gödel's incompleteness theorems showed that this cannot be done: no theory that contains arithmetic can prove its own consistency, and if it is consistent, it is not complete. Despite this blow to Hilbert's conception, proof theory turned out to be a rich area of mathematical logic and mathematics. And within proof theory the 'manipulation view' of mathematical reasoning is still very much alive.

From the perspective of everyday reasoning, however, there are reasons to question this. Most psychologists agree that reasoning involves representation. About the nature of representations and how these are employed in cognitive processes there is less consensus. Some prefer to think of representations as mental models and conceptualise cognitive processes in terms of creation and inspection of such models. Others employ a more rule-based conception of what representations are and how they function. (Resemblance with the semantic model-theoretic approach to validity in logic and the syntactic proof-theoretic one is no coincidence, of course.) Despite some obvious differences both views assume that what reasoning is about is representational, i.e., what we reason with has content that is assumed to be about something distinct from whatever representations themselves that we use. On the mental models view the reasoning as such is inherent in the representations, on the more rule-based view the rules as such are not representations, but they are applied to structures that represent. And there seems to be ample justification for that assumption: when we reason, we always reason *about something*, i.e., whatever representation or model we use, it stands for something external, different from itself, in most cases something in the outside world (although we do reason about our inner mental life as well, of course).

In this respect, the way in which we define and study logical systems may be a bit misleading when it comes to the relation with actual reasoning.[7] In order to define logic in as 'pure' a sense as possible, we strip away any non-logical content. That is fine if we want to focus purely on the logic as such. But in everyday life, as well as in science, reasoning is always *about something*, i.e., it always also involves non-logical content.

Group 3:
"Engel (231) claims that certain properties of logical constants are 'primitively obvious'. Do you agree? If so, what would be examples? If not, why not?"

An appeal to 'obviousness' should set off an alarm. It is not that some things are not obvious, or can be, or need to be, considered as such within a certain context. But in every such case that is something that needs to be argued. Simply stating that something is obvious, without providing reasons, will not do. For then such claims are impossible to argue against, let alone to refute, and we end up with a kind of Cartesian foundationalism: logical constants as 'clear and distinct ideas'.

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[5] See M. Detlefsen, 'Formalism', in: S. Shapiro (ed), *The Oxford Handbook of Philosophy of Mathematics and Logic*, Oxford: Oxford University Press.
[6] See R. Zach, 'Hilbert's Program', in: *The Stanford Encyclopedia of Philosophy* (Winter 2023 edition).
[7] Recall the arguments of Harman, in his *Change in View*, that we discussed earlier, and our evaluation of them in the discussion notes.

Is Engel guilty of that? On p. 232 he states: "The relationship between impressions of validity and validity has nevertheless to be close enough, if the meaning of logical words is supposed to be normative." I.e., at the level of the 'primitively obvious' what strikes us as correct *is* correct, and vice versa. Note that Engel links the 'obviousness' to normativity. And elsewhere in the paper Engel does provide some arguments for that kind of normativity. So, at least to some extent this is an appeal to obviousness that comes with arguments.

It is interesting to note that Peacocke uses negation as his example. That is remarkable given the fact that negation has spawned some of the best-known instances of pluralism: depending on what properties we associate with negation we get minimal logic, or intuitionistic logic, or classical logic. (But cf., above, on embeddability of classical and intuitionistic logic.) What role does 'obviousness' have to play here?

And then there are paraconsistent logics. These are logics that, unlike the other three, do not have the principle of non-contradiction: $\neg(\varphi \,\&\, \neg\varphi)$ is not a valid principle in paraconsistent logics.[8] These are logical systems that embrace rather than avoid contradictions.

Now, if someone claims that one of these logics is 'the right one', they must have a different concept of negation. And that seems difficult to square with Peacocke's claim that negation is 'primitively obvious'. So, this type of logical pluralism does not sit very well with the appeal to 'obviousness'.

Group 4:
"Does it make sense to assign different degrees of 'logicality' to different (groups of) people? If so, what criteria should we apply? If not, why not?"

It seems important to start with the observation that in many of its uses 'logical' is an evaluative term, and a positive one at that. A decision, or an argument, or an action that is called 'logical' is thereby (implicitly) positively evaluated. To say of a person that they can reason logically is likewise to describe them in a positive manner. (Of course, '*only* logical' might also be used as a negative characterisation, to indicate a lack of empathy, for instance.) This holds not just for individuals, but also for groups of people. The capacity for logical, rational thought, for rationally grounded action, is generally highly valued and is often used to distinguish between groups of people in an evaluative manner.

With that in mind, what about the question posed? One case of distinguishing between groups of people in terms of 'logicality' (understood as the capacity for rational, logic-based reasoning and decision-making) relates to cognitive development. It seems justified to distinguish young infants from adolescents and from adults in terms of their relative lack of 'logicality'. A prime example is the so-called 'Theory of Mind'. This is the ability to reason correctly about the beliefs that other people have that may differ from your own. This ability develops only around the age of four years and is often linked to the realisation of a particular level of linguistic competence.[9] People with specific neuropsychological disorders, such as various forms of autism, have also been found to have different development with regard to specific types of logical reasoning.[10]

[8] For an overview see G. Priest, K, Tanaka, & Z. Weber, 'Paraconsistent Logic, *The Stanford Encyclopedia of Philosophy* (Spring 2022 edition).
[9] There is a vast literature about this topic. See H. M. Wellman, 'Theory of Mind: The State of the Art', *European Journal of Developmental Psychology*, 15(6) for an overview.
[10] See, e.g., J. Pijnacker et.al., 'Defeasible reasoning in high-functioning adults with autism: Evidence for impaired exception-handling', *Neuropsychologica*, vol. 47(3), 2009.

A different case that has been defended is that between different socio-cultural groups. Cultural anthropologists and social psychologists have studied how different such groups perform on various reasoning tasks, and they seem to have found substantial differences. (Classic examples are the investigations of the Russian neuropsychologist Alexander Luria among isolated communities in the Caucasus, and that of psychologist Sylvia Scribner among people in Liberia.[11]) However, there is still a lot of discussion about the proper interpretation of these results. For one thing, the typical experimental set up of these investigations might appeal better to the typical Western or Chinese test populations that are usually recruited from university students, while they are quite foreign to other people. Also, the very idea that reasoning can be dissociated from content, i.e., from practical problems that need to be solved, is typically one that appeals more to people with a formal academic background than to people with more practical concerns. So, the observed differences need to be interpreted against the backgrounds of the concerns of the various groups. When that is taken into account, matters appear to be far more complex. There isn't a simple, universal measure that can be used to judge 'logicality'. Different social, economic and cultural backgrounds require often subtle modifications of the norms for judging people's ability to reason and to act 'in a logical manner'.[12]

And yet another case concerns groups on an evolutionary timescale. It seems likely that like other cognitive capacities, the capacity for logical reasoning has evolved over time. However, it is extremely difficult to come up with substantiated evidence for how that process has run its course. To judge 'logicality' of people in the past we typically need to rely on written records. Other types of evidence are scarce, if they exist at all. But written records only go back so far. The starting point of the development of writing, let alone the point in time from which written records have survived, are points in the past that are hardly relevant for judging the evolutionary development of humans in this regard.

[11] See A. R. Luria, *The Cognitive Development: Its Cultural and Social Foundations*. Harvard University Press, 1976 and S. Scribner & M. Cole, *The Psychology of Literacy*. Cambridge, Mass.: Harvard University Press, 1981.
[12] For more discussion see K. Stenning & M. van Lambalgen, *Human Reasoning and Cognitive Science*. MIT Press, Cambridge, Mass., 2008 .